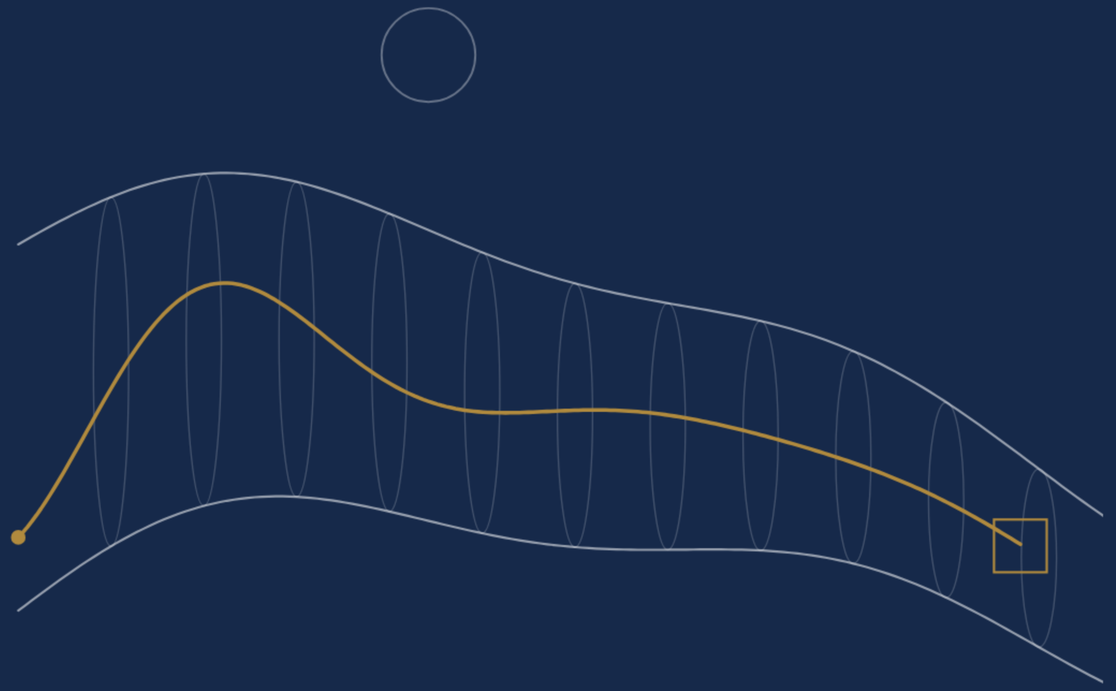
POSTDOCTORAL INTERVIEW · CONTROL & INSTRUMENTATION ENGINEERING · KFUPM

# Safe, Compliant and Adaptive Control for Robots that Interact



**Chayan Kumar Paul**
Research Associate, FOCAS Lab
Robert Bosch Centre for Cyber-Physical Systems, IISc Bangalore

4 November 2026

PROFILE

# Background

CURRENT

## Research Associate, FOCAS Lab

Robert Bosch Centre for Cyber-Physical Systems, IISc Bangalore

With Dr. Pushpak Jagtap: spatiotemporal tubes and funnel-based interaction control

PHD, 2026

## Electrical Engineering, IIT Delhi

*Safe and Compliant Adaptive Control Strategies for Human–Robot Interaction*

Advisor: Prof. Indra Narayan Kar

RESEARCH QUESTION

*How can a robot yield to people and to uncertain environments while provably never violating safety — with limited model knowledge and explicit time guarantees?*

**2** journal articles (IJSS, IJHCI)

**4** manuscripts under review

**1** IEEE conference paper (CoDIT)

**2** controllers validated on hardware

RESEARCH THEME

# Interaction that is safe by construction

## 01
### Learn

Identify unknown parameters **in fixed time**, from initial rather than persistent excitation.

*Certificate:* explicit settling-time bound $\mathcal{T}_{\max}$

## 02
### Constrain

Never leave the safe set — **even while yielding** to the human.

*Certificate:* forward invariance; recursive feasibility

## 03
### Estimate

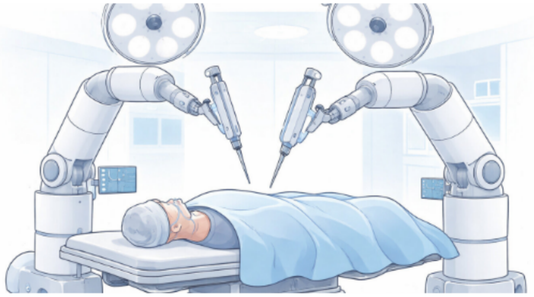
Reconstruct **unmeasured states**, treating the human force as an unknown input.

*Certificate:* contraction ⇒ exponential convergence

*Separate what must be guaranteed from what can be learned — and prove both.*

MOTIVATION

# Compliance and safety under uncertainty





TASK-SPACE DYNAMICS AND TARGET IMPEDANCE

$$M_p(q)\ddot{p} + C_p(q,\dot{q})\dot{p} + G_p(q) = J^{\dagger\top}(q)\,\tau + f_{ext}$$

$$M\ddot{e} + D\dot{e} + Ke = f_{ext}, \qquad e = p - p_d$$

| | Model-free | Parameter convergence | Explicit time | Hard safety |
|---|---|---|---|---|
| Classical impedance | × | – | – | × |
| Adaptive impedance | ✓ | PE only | × | × |
| Invariance control | × | – | – | ✓ |
| CBF / MPC based | × | – | – | ✓ |
| BLF / PPC admittance | ✓ | × | × | ✓† |
| **This work** | ✓ | ✓ | ✓ | ✓ |

† conservative: bounds tied to the barrier construction rather than the task. “Model-free”: unknown parameters with known regressor structure.

**The gap.** No single framework delivered compliance, parameter convergence in bounded time and hard state constraints — together, under model uncertainty and disturbances.

# Fixed-time adaptive impedance control

**DESIGN**

$$M(\dot{v} + \lambda v) = M\ddot{e} + D\dot{e} + Ke - f_{ext}$$

$$\tau = J^{\top}\left(\Phi\hat{\Theta} - f_{ext} - k_1\lceil v \rfloor^{1-\alpha} - k_2\lceil v \rfloor^{1+\alpha}\right)$$

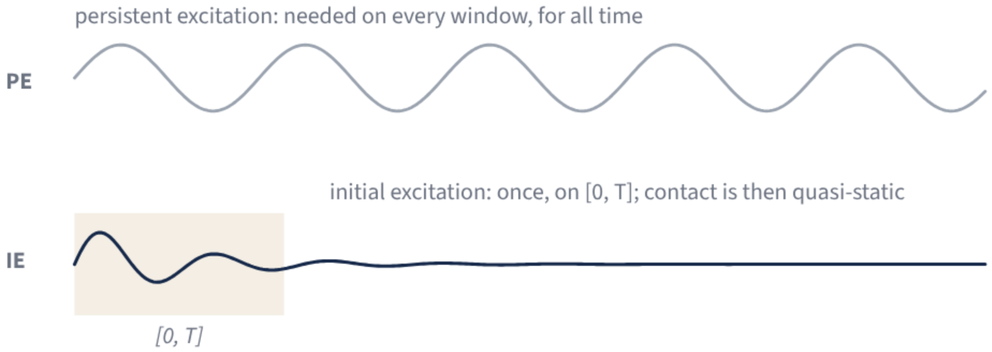
$$\dot{\hat{\Theta}} = \Gamma\Phi^{\top}v + \Gamma\sum_{j\in\{F,IF\}}\left(\lceil \varepsilon_j \rfloor^{1-\alpha} + \lceil \varepsilon_j \rfloor^{1+\alpha}\right)$$

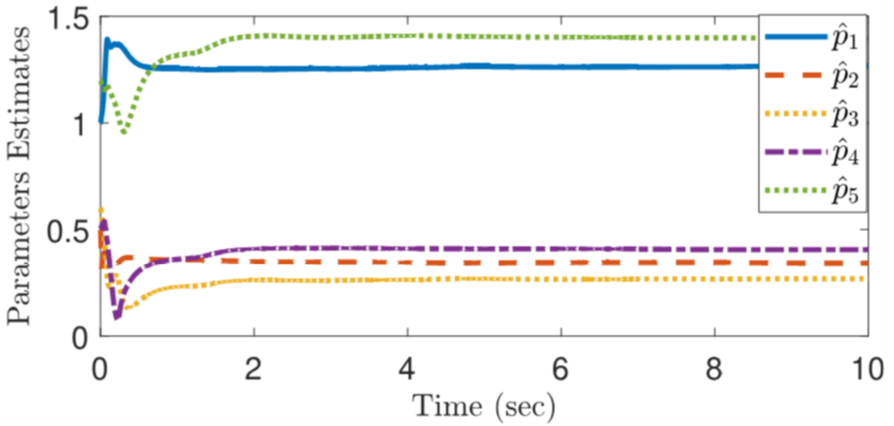
$\lceil x \rfloor^{a} = |x|^{a}\,\mathrm{sign}(x)$ elementwise, $\alpha \in (0,1)$; $\varepsilon_j$: prediction errors of filtered regressors.

**Theorem.** If the filtered regressor is initially exciting on $[0, \epsilon]$, then for every initial condition

$$v(t) = 0, \quad \tilde{\Theta}(t) = 0 \qquad \forall\, t \geq \mathcal{T}_{\max} = \epsilon + \frac{\pi}{\alpha\sqrt{\gamma_3\gamma_4}}.$$

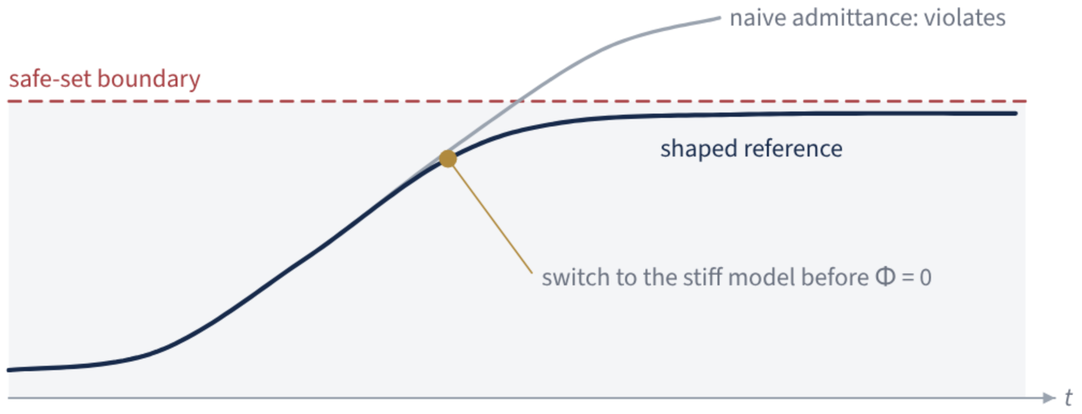
*Submitted to Int. J. Adaptive Control and Signal Processing — with D. Efimov and R. Ushirobira (Inria)*

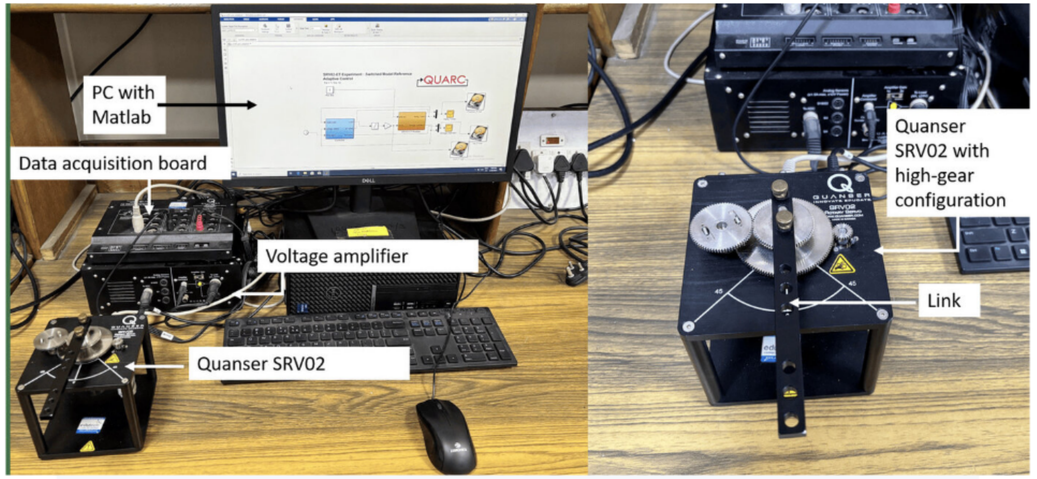



Two-link simulation: all five estimates settle well inside $\mathcal{T}_{\max} = 12.3$ s. Hardware (Quanser SRV02): $\mathcal{T}_{\max} = 39.7$ s, smooth and bounded input.

# Safety in admittance control by reference shaping



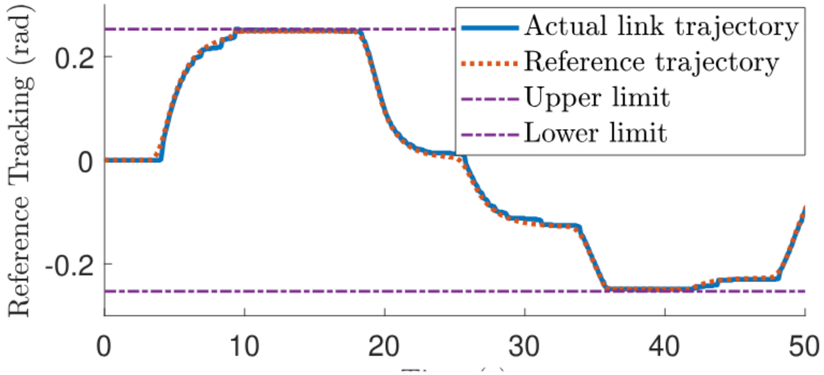




**SWITCHED REFERENCE MODEL AND INVARIANCE FUNCTION**

$$\dot{E}_r = A_\sigma E_r + B_a f_{ext}, \qquad \sigma \in \{\text{compliant, stiff}\}$$

$$\Phi(E_r) = \begin{cases} h, & \dot{h} \le 0 \\ h + \dfrac{\dot{h}^2}{2\gamma}, & \text{otherwise} \end{cases} \qquad \mathcal{G}_1 \oplus \mathcal{D} \subseteq \mathcal{S}_r$$

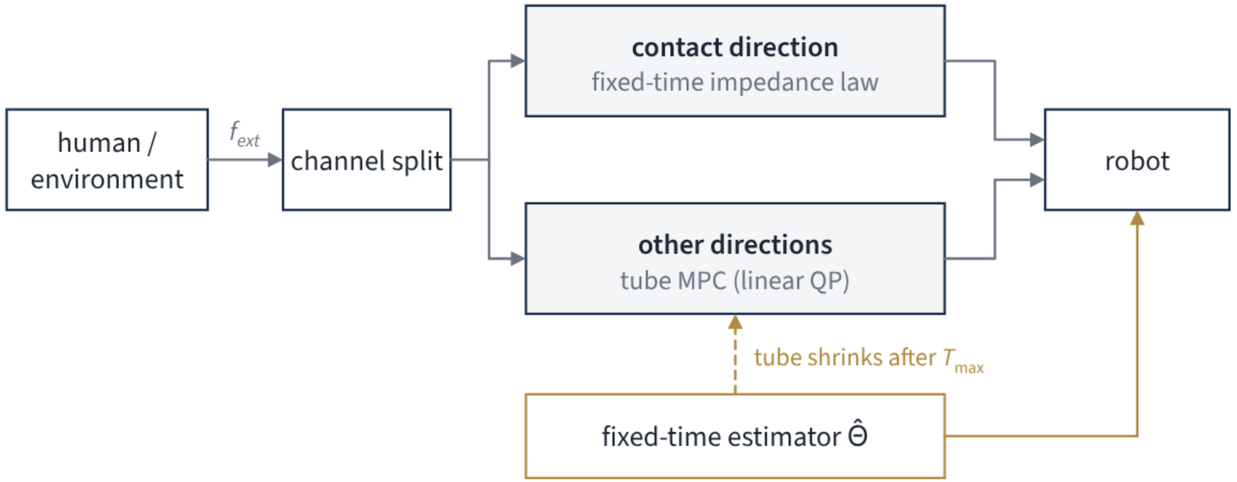
Safe set $\{h \le 0\}$; $\Phi \le 0 \Leftrightarrow$ the braking distance at deceleration $\gamma$ fits before the boundary. $\mathcal{G}_1 = \{\Phi \le 0\}$ invariant; MRAC error bound $\mathcal{D}$.

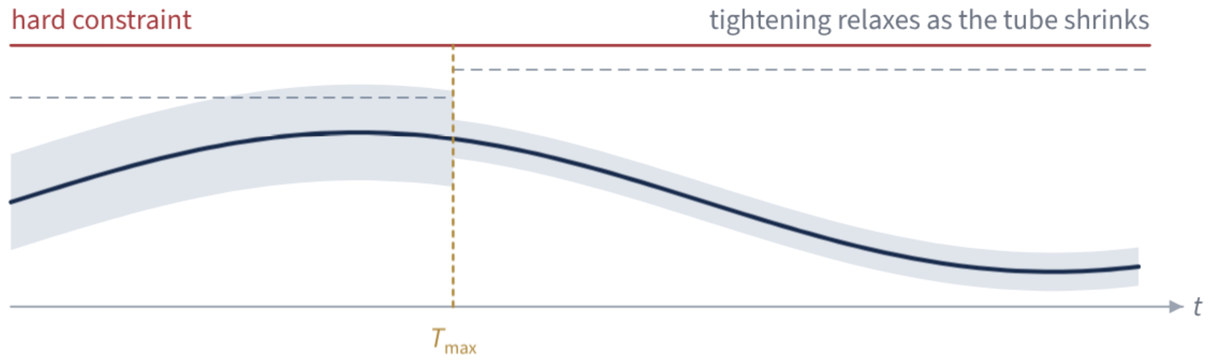


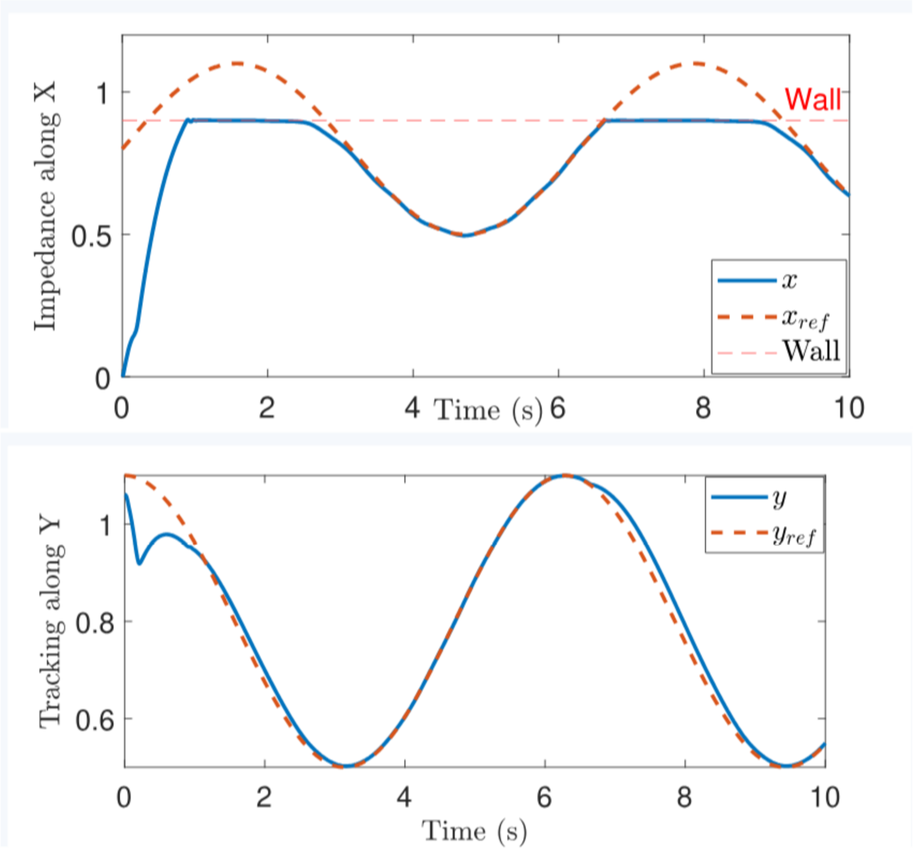

Hardware: the operator pushes the link both ways ($|f_{ext}| \lesssim 2$ N); the link complies but never crosses the limits.

*Int. J. Human–Computer Interaction (2026) · IEEE CoDIT (2024)*

# Fixed-time adaptation with tube MPC



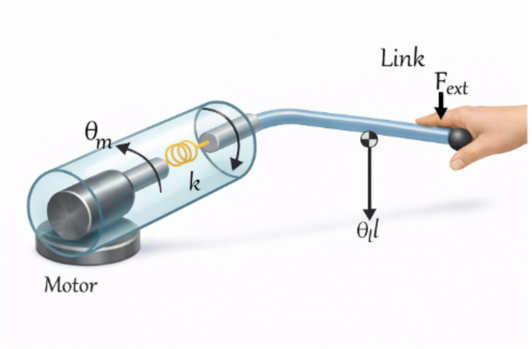




Simultaneously: compliant contact with a wall along $X$ (top) and accurate tracking along $Y$, with the $Y$-channel constraint enforced (bottom).

*In revision, IEEE Transactions on Automation Science and Engineering*

# Unknown-input dynamic observer via contraction




Flexible joints give inherent compliance, but link-side states are hard to measure and the human force is unknown.

**SYSTEM AND ANNIHILATOR**

$$\dot{x} = Ax + g(y,u) + \phi(x) + Fd, \quad y = Cx$$

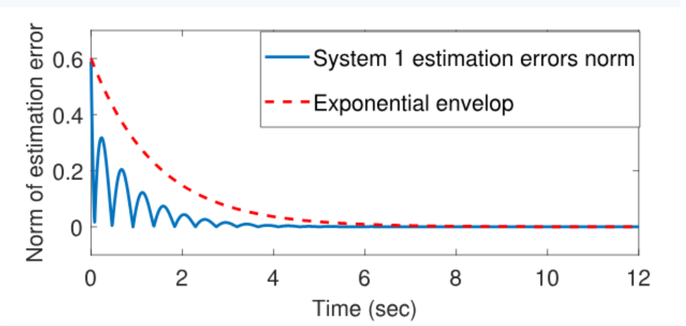
$$TF = 0 \;\Rightarrow\; T\dot{x} = TAx + Tg + T\phi(x)$$

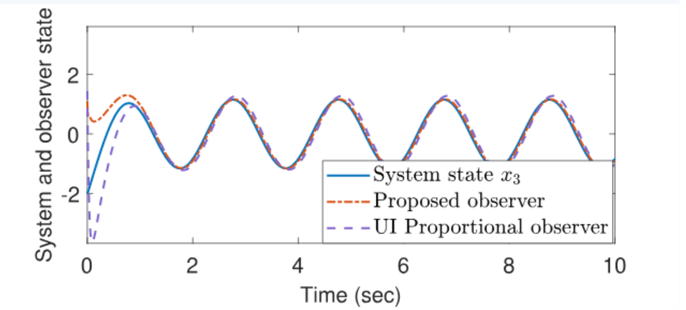
**DYNAMIC OBSERVER**

$$\dot{z} = Pz + Qy + Tg + T\phi(\hat{x}) + Rv$$

$$\dot{v} = Xz + Yy + Sv, \qquad \hat{x} = Wz + Zy$$

Exists iff $\operatorname{rank}\begin{bmatrix} T \\ C \end{bmatrix} = n$. Contraction of a virtual system, certified by an LMI, gives exponential convergence for any initial estimate.

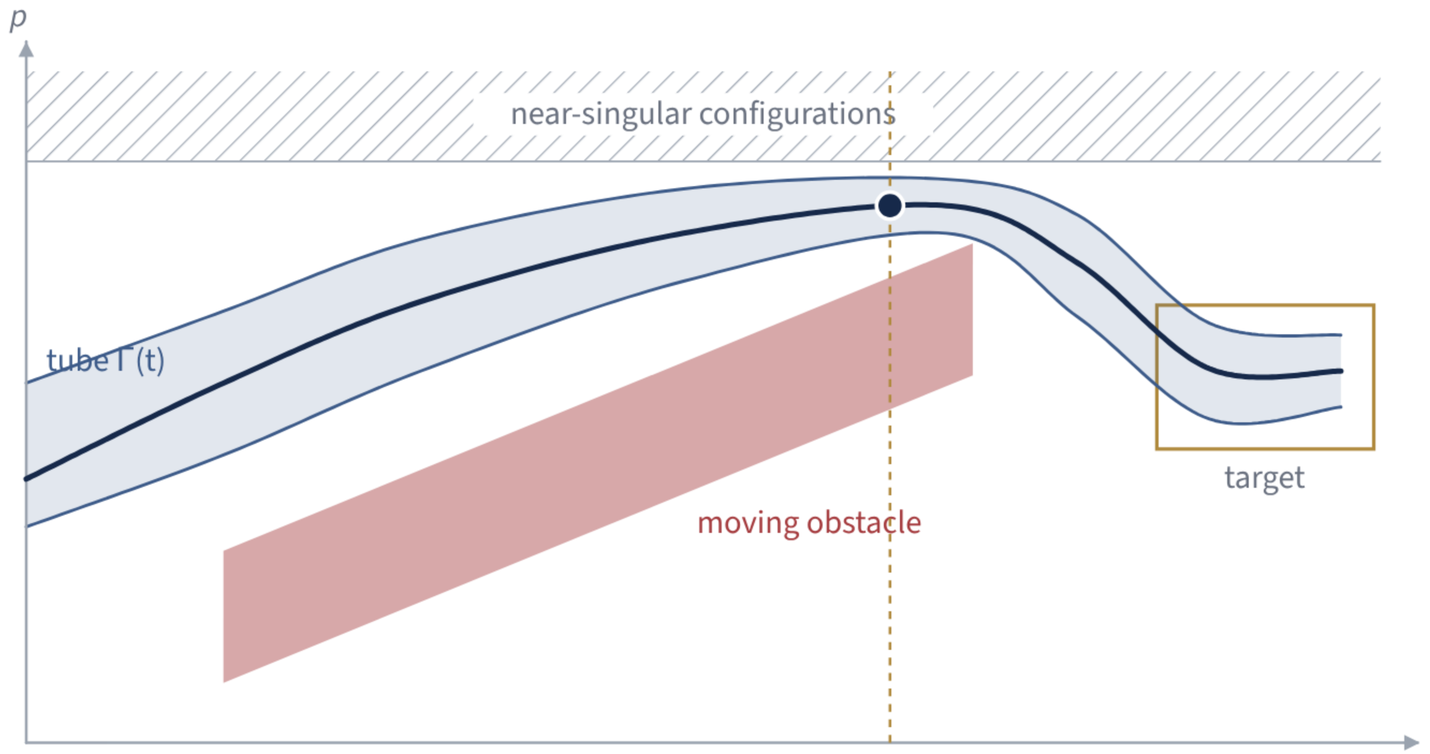


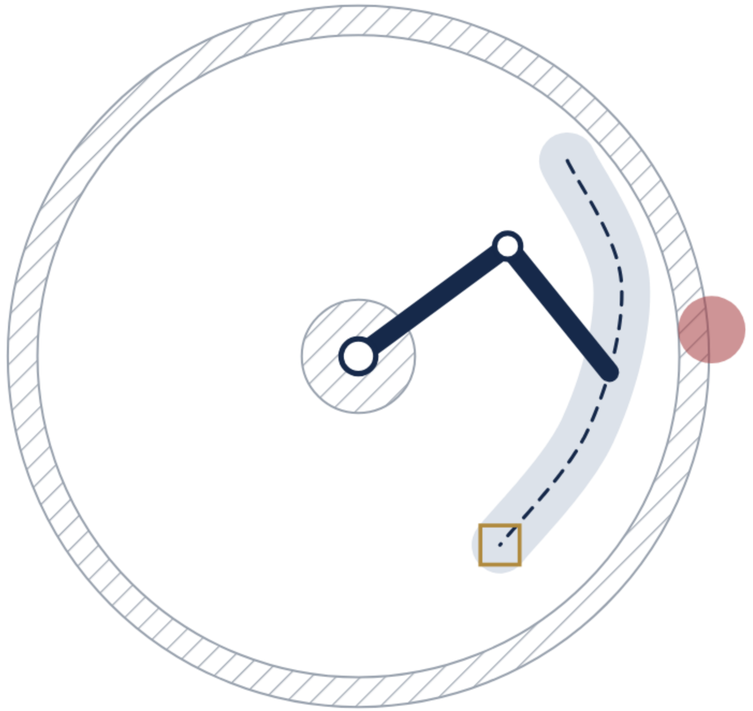




Error within an exponential envelope; faster than a proportional UI observer (simulation).

*Int. J. Systems Science (2025)*

# Spatiotemporal tubes for safe task-space control

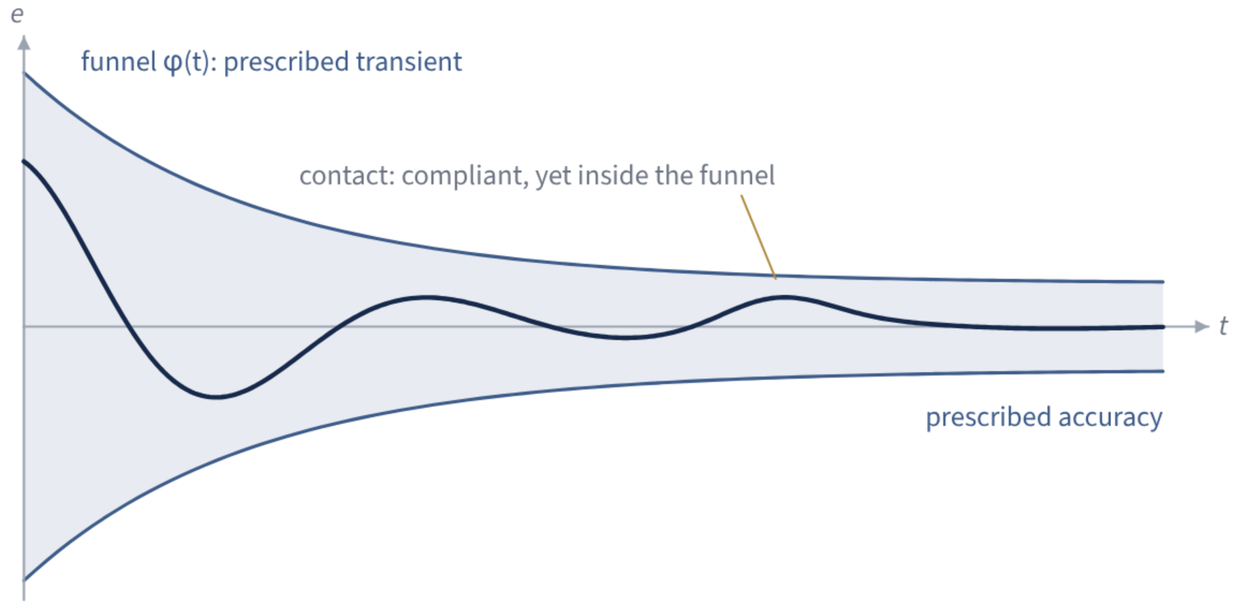


hatched: singular configurations (arm stretched / folded)

**Idea.** Shape a tube $\Gamma(t) = \{p : \gamma_L(t) < p < \gamma_U(t)\}$ with $\Gamma(t) \cap \left(\mathcal{O}(t) \cup \mathcal{S}_{\text{sing}}\right) = \emptyset$; a closed-form, approximation-free law keeps $p(t) \in \Gamma(t)$ for all $t$ — no model learning, no online optimization.

# Funnel-based interaction control




IDEA

Keep the interaction error inside a designed funnel, $\|e(t)\| < \varphi(t)$, with a low-complexity, model-free law

$$u = -\frac{k}{1 - \|e\|^2/\varphi(t)^2}\, e$$

Performance by design; no parameter estimation; the gain rises only near the funnel boundary.

ALSO IN PROGRESS

Observer-based fixed-time nested sliding-mode control of a flexible-link manipulator (with IIT Delhi).

> The same tube and funnel geometry gives **safety** (spatiotemporal tubes) and **prescribed interaction performance** (funnels); fixed-time tools give the **timing certificates**.

TRACK RECORD

# Publications

## JOURNALS

| | |
|---|---|
| **C. K. Paul**, I. N. Kar, S. Janardhanan. Unknown input dynamic observer for a class of nonlinear systems using contraction analysis. *Int. J. Systems Science*, 2025. | PUBLISHED |
| **C. K. Paul**, B. S. Dey, I. N. Kar. Safety in admittance control using reference trajectory shaping. *Int. J. Human–Computer Interaction*, 2026. | ACCEPTED |
| **C. K. Paul**, B. S. Dey, I. N. Kar. Safe physical human–robot interaction via fixed-time concurrent-learning impedance control. *IEEE T-ASE*. | IN REVISION |
| **C. K. Paul**, K. Nath, I. N. Kar, D. Efimov, R. Ushirobira. Fixed-time adaptive impedance control with excitation-aware parameter estimation. *Int. J. Adapt. Control Signal Process.* | SUBMITTED |
| A. Sharma, **C. K. Paul**, S. Janardhanan. Observer-based fixed-time nested sliding-mode control for tip-position regulation of a single-link flexible manipulator. | SUBMITTING |

## CONFERENCES

| | |
|---|---|
| **C. K. Paul**, B. S. Dey, I. N. Kar. Safe human–robot interaction using switched model reference admittance control. *IEEE CoDIT*, Malta, 2024. | PUBLISHED |
| Two further manuscripts under double-blind review (IEEE RA-L, ICRA). | UNDER REVIEW |

### METHODS

Fixed-time stability, adaptive and concurrent learning, set invariance, tube MPC, contraction, sliding modes, funnels and tubes.

### IMPLEMENTATION

Real-time experiments on the Quanser SRV02; MATLAB/Simulink, Python, ROS 2, robot simulators.

### COLLABORATION

IIT Delhi, IISc Bangalore, Inria (France).

OUTLOOK

# Research plan at KFUPM CIE

## T1
### Safe human–robot collaboration

- Tube and funnel interaction control with fixed-time adaptation on cobot arms
- Time-varying, unmodelled human dynamics

## T2
### Observer-based and fault-tolerant safety

- Safe interaction without force sensing or full state
- Tubes that survive actuator faults; discrete-time guarantees

## T3
### Safe multi-robot and aerial systems

- Tubes for fleets and formations under communication limits
- Aerial manipulation; links with CBF-based task allocation

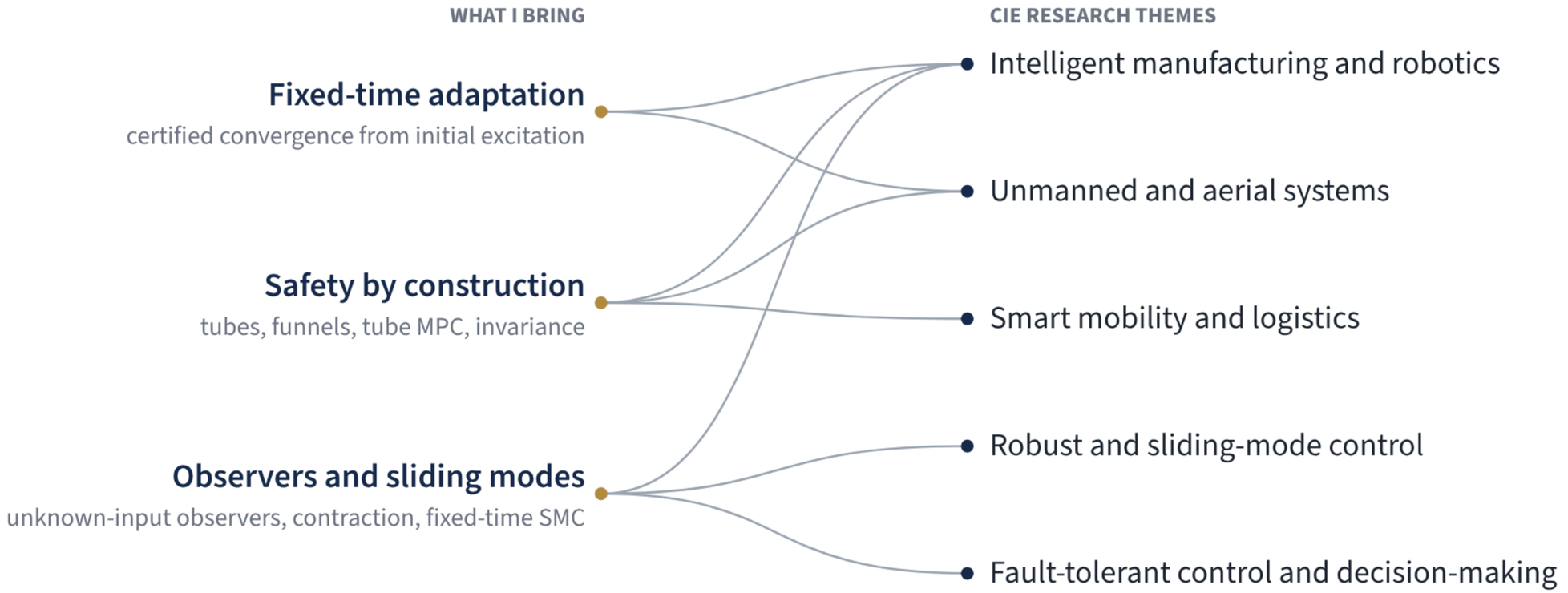
**Months 0–8**
T1 on lab cobots; T2 theory; first journal submission

**Months 8–16**
T1 experiments; T3 in simulation; conference papers

**Months 16–24**
Integration and demonstrations; joint proposals; student co-supervision

OUTLOOK

# Fit with the department




> I would bring certified safety and fixed-time learning for physical interaction, and connect them to the department's robotics, unmanned-systems and logistics projects.

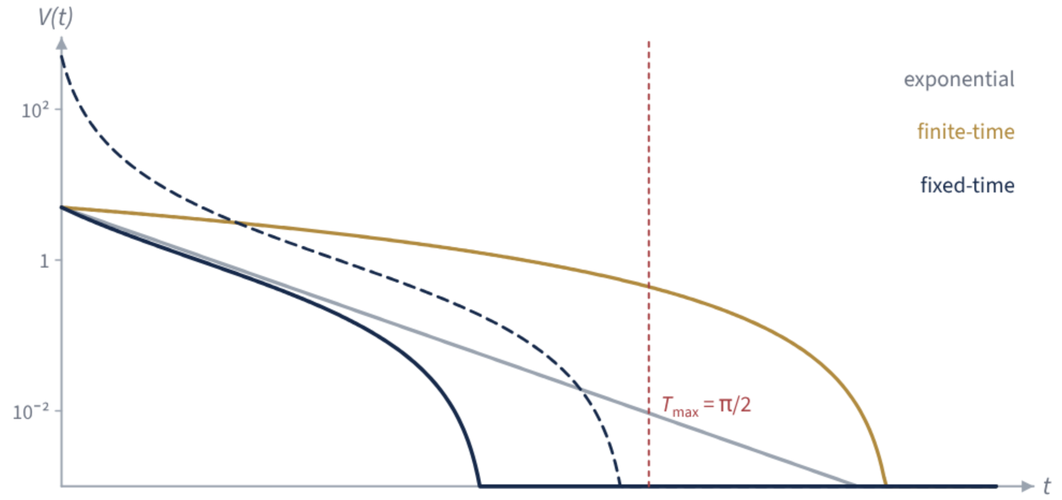
# Thank you

- Compliant, provably safe interaction control with explicit settling-time certificates
- Hardware-validated on a one-degree-of-freedom testbed; published and under review in leading journals
- A 24-month plan on certified-safe autonomy around people, aligned with CIE research

Chayan Kumar Paul · chayankumar1@iisc.ac.in | Backup slides: press ↓

BACKUP A

# Stability notions and the fixed-time bound




$\dot V = -4V\cdot\dot V = -2\sqrt{V}\cdot\dot V = -2\sqrt{V} - 2V^{3/2}$ (log scale). Dashed: fixed-time from $V_0 = 500$, still settled before $\mathcal{T}_{\max}$.

**LEMMA (POLYAKOV, IEEE TAC 2012)**

If $\dot{\mathcal{V}} \le -\gamma_a \mathcal{V}^{1+p} - \gamma_b \mathcal{V}^{1-p}$ with $0 < p < 1$, the origin is fixed-time stable and

$$\mathcal{T}(x_0) \le \mathcal{T}_{\max} = \frac{\pi}{2p\sqrt{\gamma_a \gamma_b}}.$$

Proof idea: $w = \mathcal{V}^p$ gives $\dot w \le -p(\gamma_b + \gamma_a w^2)$; the arctangent is bounded by $\pi/2$.

**IN MY DESIGNS**

$\lceil\cdot\rfloor^{1-\alpha}$ acts near the origin, $\lceil\cdot\rfloor^{1+\alpha}$ far away. The right-hand side is continuous, so there is no sliding-mode chattering. A smaller $\mathcal{T}_{\max}$ requires larger $\Gamma$, $k_1$, $k_2$ and gives higher torque peaks.

BACKUP B

# Why initial excitation is enough

REGRESSOR FILTERS (ZERO INITIAL STATES)

$$Z\Theta = \tau_1$$

$$\dot{Z}_F = -kZ_F + Z, \quad \dot{\tau}_{1F} = -k\tau_{1F} + \tau_1$$

$$\dot{Z}_{IF} = -kZ_{IF} + Z_F^\top Z_F$$

If IE holds on $[0, \epsilon]$, then $\lambda_{\min}(Z_{IF}) \geq \mu > 0$ on $[\epsilon, \epsilon_1]$ — long enough to converge before $\mathcal{T}_{\max} \leq \epsilon_1$, after which $\tilde{\Theta} = 0$ is invariant (disturbance-free case). With forgetting $k > 0$ and no new excitation, $\lambda_{\min}(Z_{IF})$ eventually decays; hence the window.

PHYSICAL VIEW

A contact task has a short transient followed by quasi-static contact: PE fails, IE holds.

PRACTICAL CHECK

Monitor $\lambda_{\min}(Z_{IF}(t))$ online; the certificate is active once it exceeds $\mu$.

SPECIAL CASE

$\alpha = 0$ recovers the classical asymptotic combined / concurrent-learning update.

BACKUP C

# Tube MPC: formulation, feasibility, computation

TUBE-MPC PROBLEM

$$\min_{\bar{u}} \ \sum_{j=0}^{N-1} \left( \|\bar{v}_{k+j}\|_Q^2 + \|\bar{u}_{k+j}\|_R^2 \right) + \|\bar{v}_{k+N}\|_P^2$$

$$\text{s.t. } \bar{v}_{k+j+1} = A_d \bar{v}_{k+j} + B_d \bar{u}_{k+j},$$

$$\bar{v}_{k+j} \in \mathcal{V} \ominus \mathcal{E}_b, \quad \bar{v}_{k+N} \in \mathcal{V}_f,$$

$$u_k = \bar{u}_k^* + K_d(v_k - \bar{v}_k), \quad A_d + B_d K_d \text{ Schur}$$

GUARANTEES

Recursive feasibility via the shifted sequence; $J^*$ is a Lyapunov function, so $\bar{v}_k \to 0$; the true state stays in $\bar{v}_k \oplus \mathcal{E}_b$, with $\mathcal{E}_b$ robust positively invariant.

LESS CONSERVATIVE

The disturbance set has a parametric part that vanishes after $\mathcal{T}_{\max}$, so the tube can be recomputed smaller.

COMPUTATION

$A_d = e^{-\lambda T_s} I$ with box constraints gives a small dense QP ($N = 40, T_s = 1$ ms). Terminal ingredients are computed offline.

IF THE BUDGET IS TIGHT

Explicit (multi-parametric) MPC, warm-started first-order solvers, or a shorter horizon with a larger terminal set.

BACKUP D

# Contraction and the observer design equations

CONTRACTION

$\dot{x} = f(x,t)$ is contracting in a metric $\mathcal{M}(x,t) \succ 0$ if

$$\frac{\partial f}{\partial x}^{\top} \mathcal{M} + \dot{\mathcal{M}} + \mathcal{M}\frac{\partial f}{\partial x} \preceq -2\beta\mathcal{M}, \quad \beta > 0.$$

All trajectories converge to each other exponentially. A virtual system with particular solutions $x$ and $\hat{x}$ then gives $\hat{x} \to x$.

LINEAR MATRIX EQUATIONS

$$PT - TA + QC = 0, \quad XT + YC = 0, \quad WT + ZC = I$$

Stack as $\Pi\Sigma = \Lambda$ with $\Pi = [P\ Q\ X\ Y\ W\ Z]$; solvable iff $\operatorname{rank}\begin{bmatrix}\Sigma \\ \Lambda\end{bmatrix} = \operatorname{rank}\Sigma$; general solution $\Pi = \Pi_0 + \mathcal{Z}_1(I - \Sigma\Sigma^{\dagger})$. The free block $\mathcal{Z}_1$ is used to satisfy the convergence LMI at the vertices bounding the Jacobian of $\phi$.

BACKUP E

# Safety tools: how they compare

| Method | Model needed | Online cost | Constraint type | Typical caveat |
|---|---|---|---|---|
| CBF–QP | control-affine model | QP each step | state set $h(x) \geq 0$ | feasibility under input limits; high relative degree |
| Barrier Lyapunov function | partial (backstepping) | closed form | error / state bounds | initial-condition requirements; design complexity |
| Prescribed performance / funnel | structural only | closed form | error envelope $\varphi(t)$ | gain grows near the boundary; noise sensitivity |
| Spatiotemporal tubes | structural only | closed form | time-varying state tube | the tube must be designed (obstacle knowledge) |
| Tube MPC | nominal linear model | QP each step | state and input sets | computation; conservatism of the invariant set |
| Reference shaping (mine) | none in the safety layer | closed form | state set via $\Phi$ | needs a good tracking loop; switching design |

BACKUP F

# Assumptions and limitations

## STANDING ASSUMPTIONS

- Dynamics linear in the parameters (known regressor)
- Interaction force measured; disturbances bounded with a known bound
- Full state available in Ch. 3–5; safe set known, initial state inside it
- Non-singular Jacobian along the task (now addressed with tubes)

## LIMITATIONS AND NEXT STEPS

- Hardware so far is single-DoF → multi-DoF cobot experiments
- Ch. 5–6 validated in simulation → hardware
- $\mathcal{T}_{\max}$ is a worst-case bound → tighter estimates
- Continuous-time proofs → discrete-time guarantees
- Observer and controller designed separately → observer-based safe interaction